

Classification in Machine Learning: How Trustworthy Classifiers Are Built

Supervised vs. Unsupervised Learning

Supervised Learning

Supervised learning is a predictive technique that trains models on labeled data. By learning a mathematical mapping from inputs to outputs, the model can accurately predict values or categories for previously unseen data.

- Relies on labeled data with known input-output pairs
- Learns relationships between features and target outcomes
- Applied to prediction and classification problems

Unsupervised Learning

Unsupervised learning operates on unlabeled data, uncovering hidden structures and patterns without explicit guidance. It can identify natural groupings within the data and serve as a foundation for downstream supervised learning tasks.

- Works with unlabeled, unannotated data
- Discovers inherent structures and patterns
- Can support subsequent prediction tasks through feature learning

Prediction Problems: Classification vs. Regression

Classification

Predicts categorical class labels — whether discrete or nominal. The model is built from a training set and its class attribute values, then applied to classify new, previously unseen data into one of the predefined categories.

Typical Applications



Credit Approval

Determining whether a customer should be approved for a loan or credit card based on their financial history and credit profile.



Fraud Detection

Classifying financial transactions as fraudulent or legitimate to protect banking and payment systems from malicious activity.

Regression

Models continuous-valued functions, enabling the prediction of unknown or missing numerical values. Unlike classification, regression operates on continuous numeric variables rather than discrete categories, making it ideal for forecasting quantities.



Medical Diagnosis

Identifying whether a tumor is malignant or benign by analyzing clinical features and patient data through a trained classification model.



Web Categorization

Assigning appropriate categories to web pages to facilitate content organization, search indexing, and information retrieval.

Classification – A Two-Step Process

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Step 1: Model Construction

The first phase involves describing a set of predetermined classes. Each tuple or sample belongs to a predefined class, determined by its class label attribute. The collection of tuples used to build the model is known as the **training set**.

- The model can be represented as classification rules, decision trees, or mathematical formulas
- Training data is used to learn patterns that generalize to unseen examples

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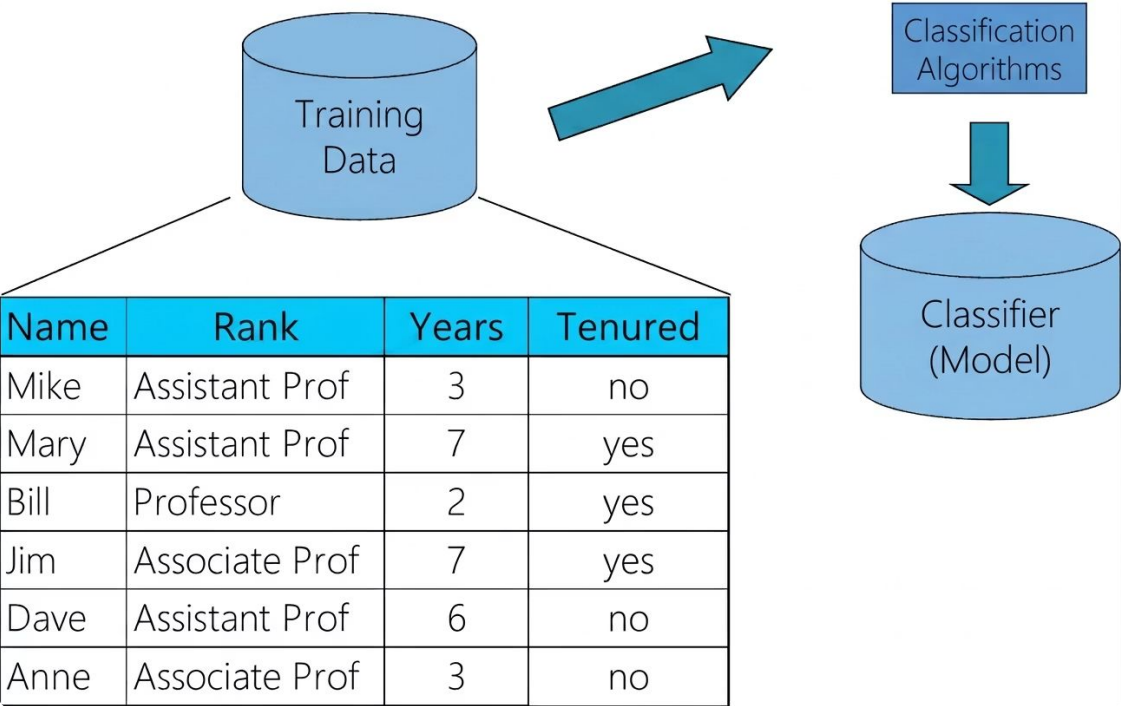
Step 2: Model Application

Once constructed, the model is applied to classify future or previously unseen objects. Its accuracy is estimated by comparing the known class labels of test samples against the model's predicted results.

- **Accuracy rate:** the percentage of test set samples correctly classified by the model
- The test set must be independent of the training set to avoid overfitting
- If the accuracy is deemed acceptable, the model can be deployed to classify new, real-world data

❏ **Important Note:** During training, if it is necessary to select among different candidate models, a portion of the training set should be set aside for evaluation — this subset is referred to as the **validation set**.

Training the Model



- **Classification Algorithms**
Process the training data to identify underlying patterns
- **Classifier (Model)**
The resulting structure capable of making predictions

Example of a Learned Rule:

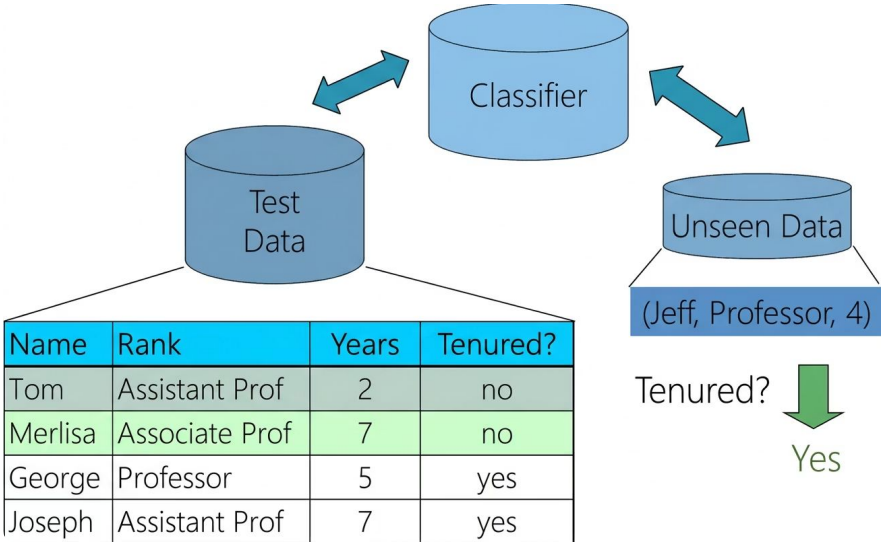
```
IF rank = 'professor'  
  OR years > 6  
THEN tenured = 'yes'
```

The training set contains labeled examples that the algorithm uses to learn patterns and relationships between features and classes.

This example illustrates how a model can learn logical rules from the training set, constructing conditions that determine the classification of new, unseen cases.

Reference: J. Han, J. Pei, and H. Tong, Data Mining: Concepts and Techniques, 4th edition. Cambridge, MA: Morgan Kaufmann, 2022.

Using the Model for Prediction



Input Data

Name: Merlisa

Role: Associate Prof

Years of Experience: 7

The model receives a new case with known features, but without an assigned class label.

Classification

The model applies the rules learned during training to determine the appropriate class.

Prediction: Tenured?

Based on the rule *IF years > 6*, the model classifies Merlisa as a candidate for tenure.

This process illustrates how the model generalizes the knowledge acquired during training to make predictions on new, unseen cases — applying the learned rules and patterns to produce accurate classifications.

Learning Roadmap

The examples are organized into three progressive groups. Study them in this order to build a solid understanding before advancing to more complex models.

Foundations

Majority baseline and Decision Tree — learn the complete cycle: data splitting, training, prediction, and evaluation.

Model Families

KNN, Naive Bayes, Random Forest, SVM, and MLP — explore different supervised learning paradigms and their classification strategies.

Model Selection

Hyperparameter search with `cla_tune` — fine-tune your models without breaking the experimental structure you have already learned.

Acknowledgements



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